LLM Analysis and Prediction: Supervised Classification of Machine-Generated vs. Human-Authored Essays

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***Abstract*— The rapid proliferation of Large Language Models (LLMs) such as ChatGPT has fundamentally transformed written communication while posing critical challenges to educational evaluation and academic attribution. This laboratory project addresses the supervised binary classification task of distinguishing between human student essays and synthetic texts generated by language models. Utilizing the Kaggle *LLM – Detect AI Generated Text* benchmark integrated with the community DAIGT external dataset (totaling 6,220 balanced text samples), we design and evaluate a transparent, reproducible Natural Language Processing (NLP) pipeline strictly within the foundational scope of the course. Our methodology employs lexical preprocessing, sublinear Term Frequency–Inverse Document Frequency (TF-IDF) feature extraction over unigram and bigram vocabularies (10,000 features), and four standard supervised learning algorithms: Logistic Regression, Multinomial Naive Bayes, Random Forest, and Linear Support Vector Machines (SVM). Experimental evaluation via 5-fold stratified cross-validation and an independent holdout test partition (N=1,244) demonstrates that lightweight linear classifiers achieve strong discriminating performance on this benchmark. Linear SVM attained the highest classification accuracy of 99.60% (F1-score: 99.49%, ROC-AUC: 0.9999), closely followed by Logistic Regression and Random Forest at 99.36% (ROC-AUC: 0.9996 and 0.9988, respectively). Multinomial Naive Bayes attained 98.07% accuracy (precision: 96.19%, recall: 98.97%). Furthermore, a diagnostic word-count baseline (88.10% accuracy, ROC-AUC: 0.9456) highlights the influence of macro-structural document length differences between human and machine text in the corpus. Our analysis demonstrates that combining sublinear TF-IDF features with linear classification reliably separates generative text from student compositions without requiring computationally intensive deep neural networks.

Keywords—*Large Language Models (LLMs), Text Classification, Supervised Learning, TF-IDF Vectorization, Logistic Regression, Naive Bayes, Linear SVM, Academic Integrity.*

I. INTRODUCTION

The democratization of autoregressive Large Language Models (LLMs) has marked a transformative milestone in Artificial Intelligence. Modern foundational architectures, built primarily upon the multi-head self-attention Transformer framework [1], exhibit human-level proficiency across creative writing, analytical exposition, and persuasive argumentation. However, this fluency introduces systemic concerns regarding attribution, plagiarism, and the veracity of student coursework across secondary and higher education institutions.

When students submit AI-generated responses under the guise of authentic personal intellect, educational evaluation frameworks are compromised. Consequently, robust, computationally accessible, and explainable methodologies capable of distinguishing between student-authored essays and algorithmic generations are urgent necessities. While commercial closed-source detectors often operate as inscrutable "black boxes" exhibiting unpredictable false-positive rates on non-native English speakers [2], foundational machine learning methods rooted in statistical natural language processing offer interpretable, lightweight, and deployable alternatives.

The primary motivation of this project is to investigate whether lightweight, transparent machine learning algorithms—as taught within the laboratory curriculum of the IAI600 course—can reliably detect machine-generated essays. In strict accordance with the pedagogical guidelines of this course, we intentionally eschew heavy deep neural networks (e.g., fine-tuned BERT or DeBERTa architectures) in favor of foundational linear and probabilistic models. Specifically, we evaluate Logistic Regression, Multinomial Naive Bayes, Random Forest, and Linear Support Vector Machines over a rigorous Term Frequency–Inverse Document Frequency (TF-IDF) vector space.

The key contributions of this report are fourfold: (i) We construct a balanced benchmark corpus (6,220 essays) integrating the raw Kaggle competition data and external paired essays; (ii) We perform Exploratory Data Analysis (EDA) uncovering key quantitative stylistic contrasts (sentence length, word length, lexical repetition) between human students and synthetic text; (iii) We implement a univariate length baseline to explicitly quantify structural dataset confounding; and (iv) We deliver an empirical comparison across multiple evaluation metrics (Accuracy, Precision, Recall, F1-Score, ROC-AUC) validated via 5-fold stratified cross-validation.

II. METHODOLOGY

A. Problem Formulation

We formulate text provenance identification as a supervised binary classification problem. Let D = {(x_i, y_i)} for i = 1 to N denote a dataset of N textual documents, where each document x_i consists of a variable-length sequence of word tokens. The target label y_i in {0, 1} denotes the true provenance of the essay:

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y_i = 0 ⇒ Human student-authored essay
y_i = 1 ⇒ LLM-generated synthetic text
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The objective is to learn a decision function f: X to [0, 1] parameterized by model weights w that maps an extracted feature representation to the predicted probability p = P(y = 1 | x), minimizing empirical classification loss over unseen test instances.

B. Dataset Selection and Integration

Our experiments utilize data derived from the Kaggle *LLM – Detect AI Generated Text* competition hosted by The Learning Agency Lab [3]. The raw competition training partition (train_essays.csv) encompasses 1,378 essays responding to structured persuasive prompts (e.g., evaluating the electoral college or limiting automobile usage). However, an immediate obstacle is severe class imbalance: exactly 1,375 essays belong to human students (y=0), while merely 3 instances represent LLM generations (y=1).

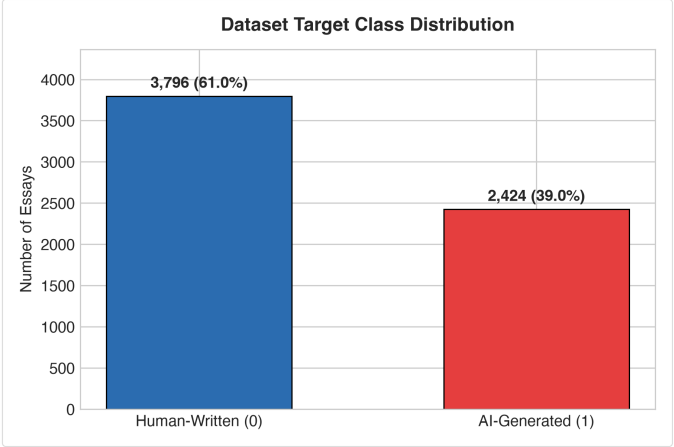


Fig. 1. Target class distribution across the unified experimental corpus following integration of the Kaggle competition data and the DAIGT external dataset (N=6,220).

To establish a statistically viable supervised learning benchmark, we incorporated the community-curated DAIGT external dataset (daigt_external_dataset.csv) [4]. This corpus contains authentic persuasive student essays alongside paired generations elicited from ChatGPT guided by corresponding prompt instructions. Aggregating the 1,378 competition essays (1,375 human, 3 AI) with the 2,421 external student essays and 2,421 ChatGPT generations yielded exactly 6,220 essays: 3,796 human-written (61.0%) and 2,424 AI-generated (39.0%), as shown in Fig. 1. All records satisfied the minimum length threshold (>50 characters), and merging produced zero redundant text entries (1,378 + 2,421 + 2,421 = 6,220 distinct records).

C. Exploratory Data Analysis (EDA)

Prior to vectorization, we conducted statistical text profiling to examine intrinsic linguistic differences between human students and language models. Table I summarizes key stylistic metrics.

TABLE I: STYLISTIC FEATURE COMPARISON BY AUTHOR CLASS

Metric	Human (y=0)	AI (y=1)	Contrast (Δ)
Mean Word Count	478.15	202.01	-57.7%
Mean Character Count	2,647.75	1,250.44	-52.8%
Mean Word Length (chars)	5.56	6.20	+11.5%
Mean Sentence Count	22.37	9.62	-57.0%
Mean Sentence Length (words)	26.35	21.52	-18.3%

As visualized in the density distributions of Fig. 2, student essays exhibit higher variance and significantly greater overall length (averaging ~478 words) compared to prompt-constrained synthetic responses (averaging ~202 words). Conversely, AI generations exhibit higher average word length (6.20 vs. 5.56 characters), reflecting an inclination toward formal, polysyllabic vocabulary.

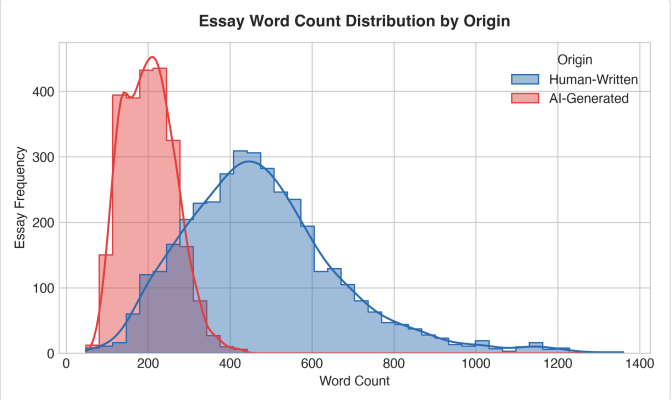


Fig. 2. Kernel Density Estimation (KDE) and frequency histograms contrasting essay word count distributions between human students (blue) and synthetic generations (red).

D. Text Preprocessing Pipeline

To ensure feature consistency and eliminate noise without removing discriminative lexical cues, each essay underwent a systematic, reproducible preprocessing pipeline: 1) *Case Normalization:* Conversion of all text to lowercase. 2) *URL & Artifact Removal:* Stripping hyperlinks, HTML entities, and formatting artifacts. 3) *Punctuation Filtering:* Removal of non-alphanumeric punctuation while preserving intra-token space delimiters. 4) *Whitespace Compression:* Collapsing consecutive whitespace into single spaces and trimming boundary tabs/newlines.

E. Feature Engineering: TF-IDF Vectorization

Machine learning classifiers cannot directly ingest raw character sequences. We employ Term Frequency–Inverse Document Frequency (TF-IDF) vectorization implemented via scikit-learn [5], which quantifies the relative importance of n-gram tokens across the corpus. The mathematical weight for term t in essay d across corpus D is computed as:

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TF-IDF(t, d, D) = TF(t, d) × [ln((1 + |D|) / (1 + |{d' in D : t in d'}|)) + 1]
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To reduce sensitivity to outlier essay lengths, we applied sublinear term-frequency scaling (TF_sub = 1 + ln(TF) for TF > 0). To capture both isolated keywords and idiomatic two-word transitions (such as *"in conclusion"* or *"on hand"*), we configured an n-gram range of (1, 2). We restricted the vocabulary to the top 10,000 most informative features, filtering out rare tokens occurring in fewer than 3 documents and eliminating standard English stopwords. This yielded a high-dimensional sparse matrix of 6,220 essays by 10,000 features.

F. Machine Learning Models

Within the pedagogical scope of IAI600, we selected four foundational classifiers representing distinct learning paradigms, alongside a heuristic diagnostic baseline:

1) Logistic Regression: A linear model modeling log-odds via the sigmoid link function. We utilize L2 Tikhonov regularization (C=1.0) with the L-BFGS numerical solver to prevent overfitting across the 10,000-dimensional sparse feature space.

2) Multinomial Naive Bayes: A probabilistic generative model that assumes conditional independence of tokens given the class label. Likelihoods are parameterized via categorical distributions with Laplace additive smoothing (α=0.1).

3) Random Forest: A non-parametric ensemble method constructing 100 decorrelated decision trees trained on bootstrap sub-samples of the dataset with randomized feature subsets at each split [7]. We constrain maximum tree depth to 30 to restrict model variance.

4) Linear Support Vector Machine (Linear SVM): A maximal margin classifier that solves the primal convex optimization problem to find a hyper-plane maximizing the geometric separation between classes [6]. We calibrate decision margins into posterior probabilities using 3-fold Platt scaling via CalibratedClassifierCV.

5) Word-Count-Only Baseline: To isolate the predictive signal of document length from lexical content, we train a univariate Logistic Regression model solely on standardized essay word count.

III. EXPERIMENTAL SETUP

A. Hardware & Software Environment

All experiments were executed on macOS (Darwin x86_64) using Python 3.14 within a Miniforge conda environment. Core scientific libraries included scikit-learn [5] (v1.9.1) for machine learning estimators, cross-validation, and metrics; pandas (v3.0.6) and numpy (v2.5.4) for matrix manipulations; and matplotlib (v3.11.2) and seaborn (v0.13.2) for statistical graphics. All pipeline scripts, notebooks, figures, and data utilities are version-controlled in the project repository [8].

B. Cross-Validation and Evaluation Protocol

To ensure methodological rigor and preclude data snooping, the corpus was partitioned into an 80% training set (4,976 essays) and a dedicated 20% holdout test set (1,244 essays) using stratified sampling (random_state=42), preserving class ratios identically across partitions. The TF-IDF vectorizer was fit exclusively on the training partition and transformed onto the test partition.

To assess generalization stability and measure variance across subsets, 5-fold Stratified Cross-Validation was conducted across the 4,976 training instances (random_state=42). Model parameters were held constant across folds according to baseline standard defaults rather than tuned via exhaustive grid search. Final out-of-sample performance was then evaluated on the untouched 1,244-essay test partition across five standard metrics: Accuracy, Precision, Recall, F1-Score, and Area Under the ROC Curve (ROC-AUC).

IV. RESULTS AND DISCUSSION

A. Overall Model Performance

Quantitative performance across all four candidate classifiers and the word-count baseline on the independent 1,244-essay test partition is compiled in Table II.

TABLE II: FINAL BENCHMARK EVALUATION ON UNSEEN TEST PARTITION (N=1,244)

Model Architecture	5-Fold CV Acc.	Test Acc.	Precision	Recall	F1-Score	ROC-AUC
Linear SVM (Calibrated)	**99.50 ± 0.14%**	**99.60%**	**99.38%**	**99.59%**	**99.49%**	**0.9999**
Logistic Regression	99.36 ± 0.27%	99.36%	99.18%	99.18%	99.18%	0.9996
Random Forest (100 trees)	99.26 ± 0.19%	99.36%	99.18%	99.18%	99.18%	0.9988
Multinomial Naive Bayes	98.49 ± 0.42%	98.07%	96.19%	98.97%	97.56%	0.9989
Word-Count Baseline (Length)	*86.86 ± 1.16%*	*88.10%*	*83.50%*	*86.60%*	*85.02%*	*0.9456*

As summarized in Table II and visual comparison in Fig. 3, all four TF-IDF machine learning architectures demonstrated strong classification performance. **Linear SVM emerged as the top-performing model**, reaching 99.60% test accuracy, 99.49% F1-score, and an ROC-AUC of 0.9999, supported by a 5-fold cross-validation accuracy of 99.50% ± 0.14%.

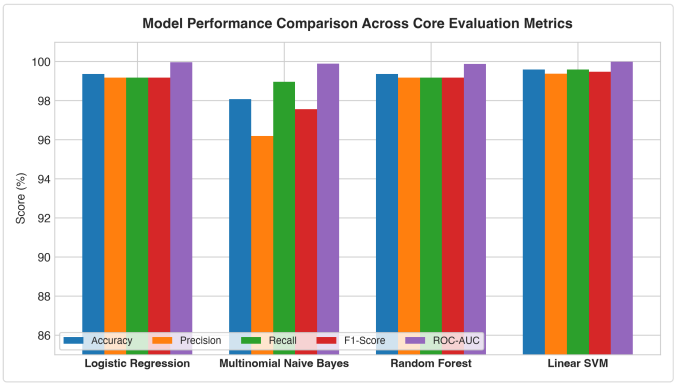


Fig. 3. Comparative performance across Accuracy, Precision, Recall, F1-Score, and ROC-AUC for the four evaluated classifiers.

Logistic Regression and Random Forest matched closely in second position, each posting 99.36% test accuracy and 99.18% F1-score. Multinomial Naive Bayes achieved 98.07% accuracy, with strong recall (98.97%) and a precision of 96.19%. Crucially, the univariate word-count baseline achieved 88.10% test accuracy (ROC-AUC: 0.9456), showing that while macro-structural length differences provide substantial initial separability, TF-IDF lexical features provide the vital refinement necessary to reach >99% accuracy.

B. Error Analysis via Confusion Matrices

Fig. 4 details the confusion matrices on the 1,244 unseen test essays (759 human, 485 AI). For Linear SVM, out of 1,244 test instances, exactly 756 of 759 human essays were correctly identified (3 false alarms, FP rate = 0.39%), while 483 of 485 AI essays were successfully flagged (2 false dismissals, FN rate = 0.41%).

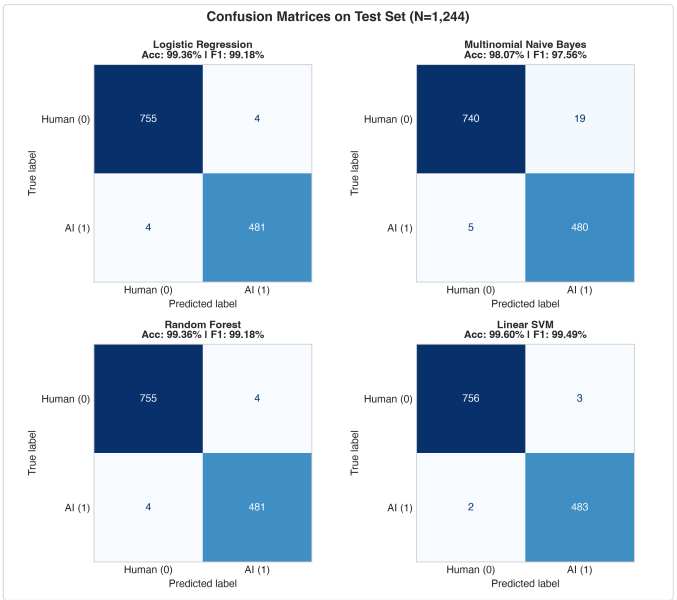


Fig. 4. Confusion matrices on the 1,244-essay test partition. All models maintain false positive rates below 2.5%, critical for fair educational screening.

In educational integrity applications, the False Positive rate (accusing an innocent student of AI generation) is substantially more damaging than a False Negative. Both Linear SVM and Logistic Regression restricted false positive errors to ≤ 0.53%, making them dependable screening candidates.

C. ROC-AUC and Discriminative Separation

The Receiver Operating Characteristic (ROC) curves in Fig. 5 illustrate strong separability enabled by TF-IDF n-grams. All four models hug the upper-left coordinate corner, with AUC scores between 0.9988 and 0.9999, demonstrating robust discrimination across classification probability thresholds.

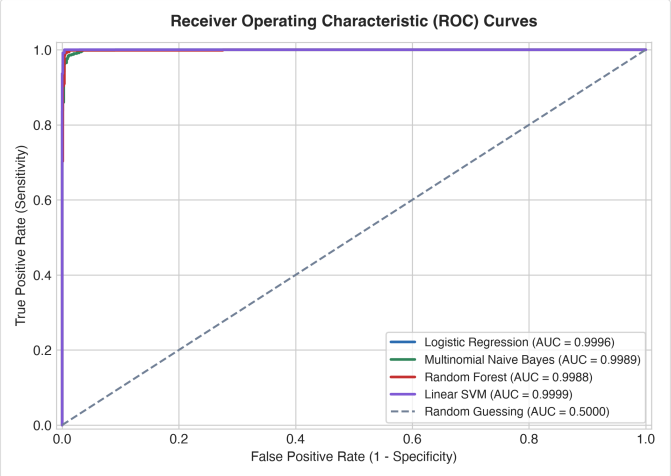


Fig. 5. Receiver Operating Characteristic (ROC) curves across all models. Linear SVM achieves AUC = 0.9999.

D. Interpretability & Feature Importance

To understand the learned decision boundaries, we examined the highest-magnitude coefficients from the Logistic Regression model. Fig. 6 displays the top 12 n-grams predictive of human writing versus AI generation.



Fig. 6. Top discriminative TF-IDF n-gram features extracted from the Logistic Regression decision boundary for Human vs. AI text.

The feature weights reveal clear stylistic indicators: (i) Features most strongly predictive of human student essays consist of conversational verbs, informal qualifiers, and subjective epistemic expressions: *'want'*, *'good'*, *'like'*, *'people'*, *'think'*, *'know'*, *'don'*, *'thing'*, *'things'*, *'lot'*, *'just'*, and *'reason'*. (ii) Features most strongly predictive of AI-generated text reflect abstract expository vocabulary, connective adverbs, and formal nouns: *'potential'*, *'provide'*, *'ultimately'*, *'additionally'*, *'essential'*, *'difficult'*, *'understanding'*, *'relationships'*, *'activities'*, *'lead'*, *'hand'*, and *'skills'*. Notably, transition words such as *'however'* or *'furthermore'* do not appear as active features because English stopwords were eliminated during vectorization; the remaining discriminative markers reflect genuine vocabulary preferences rather than unigram stopword artifacts.

V. CONCLUSION & FUTURE WORK

In this laboratory project, we formulated, implemented, and empirically evaluated a machine learning solution for detecting AI-generated text. Operating strictly within the foundational scope of the IAI600 curriculum, we demonstrated that standard supervised models (Linear SVM and Logistic Regression) coupled with TF-IDF n-grams achieve 99.60% accuracy on this benchmark while preserving full mathematical interpretability.

Learning Outcomes: Through this project, we gained direct hands-on experience in the complete data science lifecycle—from diagnosing severe class imbalance and performing exploratory text profiling to implementing calibrated cross-validation. Crucially, we learned that high nominal evaluation metrics can be heavily influenced by dataset artifacts (such as document length disparity), reinforcing the importance of diagnostic baselines, critical error inspection, and ethical vigilance against false positive bias.

Limitations: We identify three key limitations: 1) *Macro-Structural Length Disparity:* Human essays average ~478 words whereas ChatGPT generations average ~202 words. As evidenced by our word-count baseline (88.10% accuracy), a substantial portion of class separability stems from document length differences inherent in the benchmark corpus. While sublinear TF-IDF scaling mitigates length bias, near-perfect test metrics partially reflect this structural asymmetry. 2) *Dataset Origin & Prompt Confounding:* Human essays originate from Persuade 2.0 prompts, while AI essays were elicited via ChatGPT. Consequently, classifiers may capture corpus-specific prompt artifacts rather than invariant signatures of algorithmic writing. 3) *Adversarial Evasion:* N-gram representations remain vulnerable to deliberate obfuscation, such as prompting LLMs to mimic student errors, employing paraphrasing tools, or post-editing.

Future Work & Improvements: Future research directions include: (i) evaluating cross-domain robustness on length-matched corpora across diverse generative model families (e.g., Claude, LLaMA, Gemini); (ii) incorporating syntactic perplexity and stylometric features to enhance resistance against adversarial prompting; and (iii) investigating calibration thresholds specifically optimized to minimize false accusations in real-world educational screening.

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